

CAMBRIDGE INTERNATIONAL EXAMINATIONS

Cambridge International Advanced Subsidiary and Advanced Level

MARK SCHEME for the October/November 2014 series

9702 PHYSICS

9702/43

Paper 4 (A2 Structured Questions), maximum raw mark 100

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- 1 (a) (i) *either* $\omega = 2\pi / T$ *or* $\omega = 2\pi f$ and $f = 1 / T$ C1
 $\omega = 2\pi / 0.30$
 $= 20.9 \text{ rad s}^{-1}$ (accept 2 s.f.) A1 [2]
- (ii) kinetic energy $= \frac{1}{2} m \omega^2 x_0^2$ *or* $v = \omega x_0$ and $\frac{1}{2} m v^2$ C1
 $= \frac{1}{2} \times 0.130 \times 20.9^2 \times (1.5 \times 10^{-2})^2 = 6.4 \times 10^{-3} \text{ J}$ A1 [2]
- (b) (i) as magnet moves, flux is cut by cup / aluminium giving rise to induced e.m.f. (in cup) B1
induced e.m.f. gives rise to currents and heating of the cup B1
thermal energy derived from oscillations of magnet so amplitude decreases B1
or
induced e.m.f. gives rise to currents which generate a magnetic field (B1)
the magnetic field opposes the motion of the magnet so amplitude decreases (B1) [3]
- (ii) *either* use of $\frac{1}{2} m \omega^2 x_0^2$ and $x_0 = 0.75 \text{ cm}$ *or* x_0 is halved so $\frac{1}{4}$ energy C1
to give new energy = 1.6 mJ
either loss in energy = $6.4 - 1.6$ *or* loss = $\frac{3}{4} \times 6.4$ giving loss = 4.8 mJ A1 [2]
- (c) $q = mc\Delta\theta$
 $4.8 \times 10^{-3} = 6.2 \times 10^{-3} \times 910 \times \Delta\theta$ C1
 $\Delta\theta = 8.5 \times 10^{-4} \text{ K}$ A1 [2]
- 2 (a) smooth curve with decreasing gradient, not starting at $x = 0$ M1
end of line not at $g = 0$ or horizontal A1 [2]
- (b) straight line with positive gradient M1
line starts at origin A1 [2]
- (c) sinusoidal shape B1
only positive values and peak/trough height constant B1
4 'loops' B1 [3]
- 3 (a) initially, $pV/T = (2.40 \times 10^5 \times 5.00 \times 10^{-4}) / 288 = 0.417$ M1
finally, $pV/T = (2.40 \times 10^5 \times 14.5 \times 10^{-4}) / 835 = 0.417$ M1
ideal gas because pV/T is constant A1 [3]
(allow 2 marks for two determinations of V/T and then 1 mark for V/T and p constant, so ideal)

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(b) (i) work done $= p\Delta V$
 $= 2.40 \times 10^5 \times (14.5 - 5.00) \times 10^{-4}$
 $= 228 \text{ J (ignore sign, not 2 s.f.)}$ C1
A1 [2]

(ii) $\Delta U = q + w = 569 - 228$
 $= 341 \text{ J}$ M1
increase A1 [2]

- 4 (a) acceleration / force proportional to displacement (from a fixed point)
either acceleration and displacement in opposite directions
or acceleration always directed towards a fixed point M1
A1 [2]

(b) (i) zero & 0.625 s or 0.625 s & 1.25 s or 1.25 s & 1.875 s or 1.875 s & 2.5 s A1 [1]

(ii) 1. $\omega = 2\pi / T$ and $v_0 = \omega x_0$ C1
 $\omega = 2\pi / 1.25$
 $= 5.03 \text{ rad s}^{-1}$ C1
 $v_0 = 5.03 \times 3.2$
 $= 16.1 \text{ cm s}^{-1}$ (allow 2 s.f.) A1 [3]

2. $v = \omega \sqrt{(x_0^2 - x^2)}$
either $\frac{1}{2}\omega a = \omega \sqrt{(x_0^2 - x^2)}$ or $\frac{1}{2} \times 16.1 = 5.03 \sqrt{(3.2^2 - x^2)}$ C1
 $x_0^2 / 4 = x_0^2 - x^2$ $2.58 = 3.2^2 - x^2$
 $x = 2.8 \text{ cm}$ $x = 2.8 \text{ cm}$ A1 [2]

- (c) sketch: loop with origin at its centre
correct intercepts & shape based on (b)(ii) M1
A1 [2]

- 5 (a) work done / energy in moving unit positive charge
from infinity (to the point) M1
A1 [2]

(b) (i) $V = q / 4\pi\epsilon_0 r$
at 16 kV, $q = 3.0 \times 10^{-8} \text{ C}$
 $r = (3.0 \times 10^{-8}) / (4\pi \times 8.85 \times 10^{-12} \times 16 \times 10^3)$ C1
 $= 1.69 \times 10^{-2} \text{ m (allow 2 s.f.)}$ A1 [2]
(allow any answer which rounds to 1.7×10^{-2})

(ii) energy is / represented by area 'below' line
energy $= \frac{1}{2}qV$ C1
 $= \frac{1}{2} \times 24 \times 10^3 \times 4.5 \times 10^{-8}$ C1
 $= 5.4 \times 10^{-4} \text{ J}$ A1 [3]

(c) $V = q / 4\pi\epsilon_0 r$ and $E = q / 4\pi\epsilon_0 r^2$ giving $Er = V$ B1
 $2.0 \times 10^6 \times 1.7 \times 10^{-2} = V$ C1
 $V = 3.4 \times 10^4 \text{ V}$ A1 [3]

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- 6 (a) for the two capacitors in parallel, capacitance = $96 \mu\text{F}$
for complete arrangement, $1/C_T = 1/96 + 1/48$
 $C_T = 32 \mu\text{F}$ C1
A1 [2]
- (b) p.d. across parallel combination is one half p.d. across single capacitor
total p.d. = 9 V C1
A1 [2]
- 7 (a) *either* charge exists in discrete and equal quantities
or multiples of elementary charge $e/1.6 \times 10^{-19} \text{ C}$ B1 [1]
- (b) (i) force due to magnetic field must be upwards
B-field into the plane of the paper B1
B1 [2]
- (ii) sketch showing: deflection consistent with force in (b)(i)
reasonable curve B1
B1 [2]
- 8 (a) discrete amount/packet/quantum of energy
of electromagnetic radiation/EM radiation M1
A1 [2]
- (b) (i) $E = hc/\lambda$
 $= (6.63 \times 10^{-34} \times 3.0 \times 10^8)/(570 \times 10^{-9}) = 3.49 \times 10^{-19} \text{ J}$ A1 [1]
- (ii) 1. number = $(2.7 \times 10^{-3})/(3.5 \times 10^{-19})$
 $= 7.7 \times 10^{15}$ C1
A1 [2]
2. momentum of photon = h/λ
 $= (6.63 \times 10^{-34})/(570 \times 10^{-9})$
 $= 1.16 \times 10^{-27} \text{ kg m s}^{-1}$ C1
C1
- change in momentum = $1.16 \times 10^{-27} \times 7.7 \times 10^{15}$
 $= 8.96 \times 10^{-12} \text{ kg m s}^{-1}$ A1 [3]
- (allow $E = pc$ route to 9×10^{-12})
- (c) pressure = (change in momentum per second)/area
 $= (8.96 \times 10^{-12})/(1.3 \times 10^{-5})$
 $= 6.9 \times 10^{-7} \text{ Pa}$ C1
A1 [2]
- 9 (a) activity = $(1.7 \times 10^{14})/(2.5 \times 10^6)$
 $= 6.8 \times 10^7 \text{ Bq kg}^{-1}$ A1 [1]
- (b) (i) energy released per second in 1.0 kg of steel
 $= 6.8 \times 10^7 \times 0.067 \times 1.6 \times 10^{-13}$
 $= 7.3 \times 10^{-7} \text{ J}$ B1 [1]

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(ii) this is a very small quantity of energy so steel will not be warm B1 [1]

(iii) $A = A_0 e^{-\lambda t}$ and $\lambda t_{1/2} = \ln 2$ C1
 $400 = (6.8 \times 10^7) \exp(-[\ln 2 \times t]/92)$ C1
 $t = 1600$ years A1

or

$A = A_0 / 2^n$ (C1)
 $n = 17.4$ (C1)
 $t = 17.4 \times 92 = 1600$ years (A1) [3]

Section B

10 (a) (i) thermistor/thermocouple B1 [1]

(ii) quartz crystal/piezoelectric crystal or transducer/microphone B1 [1]

(b) (i) $V_{OUT} = -5V$ A1
inverting input is positive or V_- is positive or $V_- > V_+$ so V_{OUT} is negative B1
op-amp has very large/infinite gain and so saturates B1 [3]

(ii) sketch: V_{OUT} switches from (+) to (–) when V_{IN} is zero B1
 V_{OUT} is +5V or –5V M1
 V_{OUT} is negative when V_{IN} is positive (or v.v.) A1 [3]

11 (a) product of density and speed M1
density of medium, speed of wave in medium A1 [2]
(not “speed of light”, 0/2)

(b) (i) $\alpha = (6.4 - 1.7)^2 / (6.4 + 1.7)^2$ C1
 $= 0.34$ A1 [2]

(ii) $I/I_0 = e^{-\mu x}$ C1
 $= \exp(-23 \times 3.4 \times 10^{-2})$ C1
 $= 0.46$ A1 [3]

(iii) $I_R/I = (0.46)^2 \times 0.34$ C1
 $= 0.072$ A1 [2]

12 (a) analogue: continuously variable B1
digital: two/distinct levels only or 1 s and 0 s or highs and lows B1 [2]

(b) (i) 5 A1 [1]

(ii) 1 1 0 1 A1 [1]

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- (c) greater number of voltage/signal levels B1
 smaller step heights in reproduced signal B1
 smaller voltage/signal changes can be seen B1 [3]

- 13 (a) same carrier frequencies can be re-used M1
 but not in neighbouring cells/possible to use more handsets A1 [2]

- (b) e.g. wavelength is short (M1)
 so aerial on mobile phone conveniently short (A1)
 e.g. limited range (M1)
 so low power/less interference between cells (A1)
 e.g. large number of channels/greater bandwidth (M1)
 so more simultaneous callers (A1) [4]